

Anonymized Forensic Report: Bayesian Attribution Model

Scope. This report summarizes Bayesian attribution results using anonymized dataset names and anonymized candidate labels (Cluster A, Cluster B, ...).

Executive Summary

Using the supplied binomial/Bernoulli model and uniform priors over **90** candidates, **Cluster A** is the highest-probability candidate.

One-vs-many posterior (normalized across all candidates): $P(\text{Cluster A} \mid D) = 1$.

One-vs-rest posterior (candidate vs baseline non-responsibility): $P(\text{Cluster A} \mid D) = 1$.

Data Summary (Anonymized)

Evidence was derived from three anonymized sources: an IP-infrastructure dataset, an executable/hash dataset, and an overlap/behavioral signal derived from co-occurrence of infrastructure and executable evidence within a candidate.

Observed totals: **n_ip** = **986** unique IP indicators, **n_hash** = **32** unique executable/hash indicators, **M** = **90** candidate clusters.

Method

For each candidate cluster i , the likelihood is the product of: (1) a binomial model over matched IP indicators, (2) a binomial model over matched executable/hash indicators, and (3) a Bernoulli behavioral-overlap indicator.

One-vs-many posterior:

$$P(H_i \mid D) = \frac{L_i \pi_i}{\sum_{j=1}^M L_j \pi_j}$$

One-vs-rest posterior:

$$P(H_i \mid D) = \frac{L_1(i) P(H_i)}{L_1(i) P(H_i) + L_0(i) P(\neg H_i)}$$

Model Parameters (Plug-in)

Responsible-hypothesis parameters: $p1_ip=0.5$, $p1_hash=0.4$, $p1_beh=0.9$. Baseline (non-responsible) parameters: $p0_ip=0.05$, $p0_hash=0.02$, $p0_beh=0.1$.

Results

Rank	Candidate	k_ip	k_hash	beh	P(one-vs-many)
1	Cluster A	901	28	1	1
2	Cluster B	1	0	0	1.00932e-122
3	Cluster C	1	0	0	1.00932e-122
4	Cluster D	1	0	0	1.00932e-122

5	Cluster E	1	0	0	1.00932e-122
6	Cluster F	1	0	0	1.00932e-122
7	Cluster G	1	0	0	1.00932e-122
8	Cluster H	1	0	0	1.00932e-122
9	Cluster I	1	0	0	1.00932e-122
10	Cluster J	1	0	0	1.00932e-122

Interpretation

The posterior mass concentrates on **Cluster A** because it contains a large fraction of observed IP and executable indicators relative to other candidates under the plug-in parameterization. Under this model, other candidates receive negligible posterior weight.

Anonymization Notes

This document omits raw indicators (e.g., IP strings, hashes), removes local file paths, and replaces internal identifiers (e.g., component_id values) with rank-based labels (Cluster A, Cluster B, ...). Dataset names are generalized.

Limitations

Results are conditional on the independence assumptions and on the selected plug-in probabilities. If the chosen probabilities do not reflect the true base rates (especially for rare indicators), the posteriors may be miscalibrated. Consider sensitivity analysis or hierarchical (Beta-Binomial) extensions for more robust uncertainty handling.